

Mitigating Cyber and Motion Sickness with Haptic and Visual Feedback for VR Users in Autonomous Vehicle

Hyemin Shin
Digital Experience Laboratory
Korea University
Seoul, Republic of Korea
mini9974@korea.ac.kr

Yuki Shimizu
Interactive Media Design Laboratory
Nara Institute of Science and
Technology
Ikoma, Japan
shimizu.yuki.sy6@is.naist.jp

Hanseob Kim
Department of Artificial Intelligence
Konyang University
Daejeon, Republic of Korea
khseob0715@konyang.ac.kr

Taishi Sawabe*
Interactive Media Design Laboratory
Nara Institute of Science and
Technology
Ikoma, Japan
t.sawabe@is.naist.jp

Gerard Jounghyun Kim*
Digital Experience Laboratory
Korea University
Seoul, Republic of Korea
gjkim@korea.ac.kr

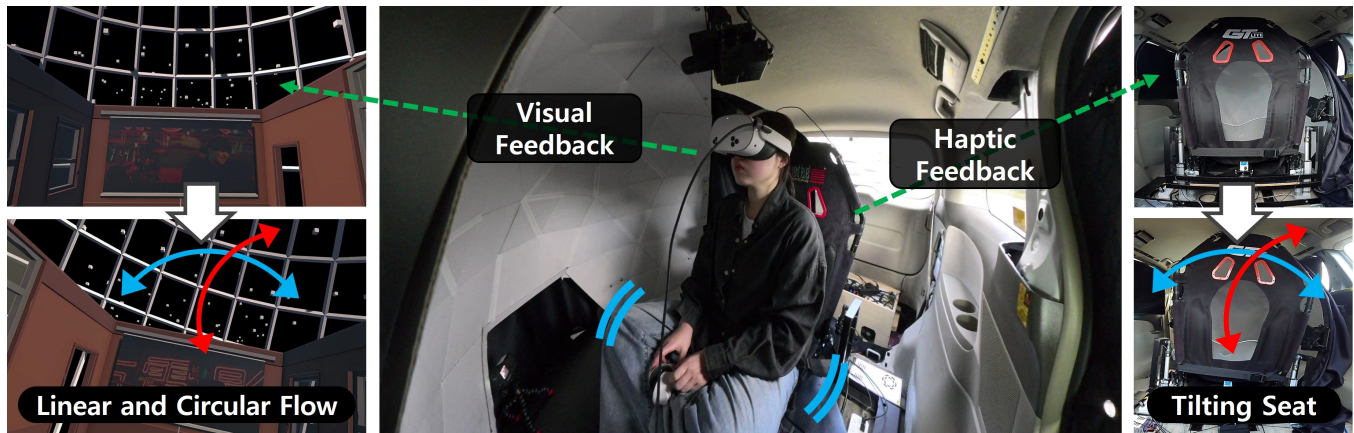


Figure 1: Overview of the experimental setup. Participants experienced passive and stationary VR content, specifically watching a music video, while seated in an autonomous vehicle. To mitigate cyber and motion sickness, both visual and haptic feedback were provided in synchronization with the vehicle's movements. Visual feedback was delivered through linear and circular flow, while haptic feedback was provided via a tiltable seat that moved in sync with the vehicle

Abstract

Experiencing VR inside autonomous vehicles can cause both VR-induced cybersickness and vehicle-induced motion sickness, which negatively affects user experience. This study investigates a mitigation approach by providing synchronized visual and haptic feedback to counteract the effects of such sickness. Specifically, visual feedback reflects the linear and rotational motion of the vehicle to the VR content using flying particles and rotating grid in the

background. The haptic feedback, compensating for the vehicle's acceleration, is delivered through a tiltable seat, according to the vehicle's motion. A user study was conducted to evaluate the effectiveness of each feedback individually and their combined effect on reducing overall sickness. Additionally, the study examined the impact on user comfort and presence. The results indicated that the visual feedback enhanced the sense of movement, while haptic feedback improved comfort. Moreover, when the visual and haptic feedback were combined, a marked and synergistic decrease in sickness was observed.

*Both authors are corresponding authors and contributed equally to this research.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '25, Yokohama, Japan

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3720076>

CCS Concepts

• **Human-centered computing** → **Virtual reality**; **Empirical studies in HCI**; • **Applied computing**;

Keywords

Cybersickness, Motion Sickness, In-Vehicle VR, User Comfort

ACM Reference Format:

Hyemin Shin, Yuki Shimizu, Hanseob Kim, Taishi Sawabe, and Gerard Jounghyun Kim. 2025. Mitigating Cyber and Motion Sickness with Haptic and Visual Feedback for VR Users in Autonomous Vehicle. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 9 pages. <https://doi.org/10.1145/3706599.3720076>

1 INTRODUCTION

Advancements in autonomous vehicle technology are opening new possibilities for passengers to engage in activities beyond simply riding in a vehicle [17, 28, 31]. While traditional activities such as watching movies, playing games, and listening to music will remain popular, new forms of activity tailored to vehicles' unique interior spaces will likely emerge [1, 7, 8]. For instance, vehicles could be used for yoga or meditation classes or transformed into mobile experience centers or performance venues [1, 8, 24]. The integration of virtual reality (VR) platforms further broadens these possibilities [12, 24]. Virtual environments inherently transcend physical constraints [25], enabling diverse social activities such as lectures, meetings, and collaborative events without requiring face-to-face interactions [3, 25, 26]. Thus, utilizing VR platforms within autonomous vehicles could allow activities that were previously constrained by limited space to be performed [1, 8]. The synergistic effects of this integration are substantial, as evidenced by on-going research into in-vehicle VR applications [5, 7, 14, 15, 21, 22]. This convergence has the potential to redefine in-vehicle experiences, introducing new paradigms for mobility and leisure.

However, the use of VR platforms in autonomous vehicles faces significant challenges due to the onset of *Mixed Sickness*, a condition stemming from two distinct sources of sickness [4, 13, 17, 18]. First, cybersickness is primarily caused by sensory conflicts between visual stimuli and the vestibular system [4, 10]. For instance, users may visually perceive motion, such as forward movement/rotation, while their body remains physically stationary, leading to a sensory conflict. Second, motion sickness is induced by the vibrations and movements of the autonomous vehicle, including irregular accelerations, decelerations, and turns. These vehicular motions disrupt the passenger's sense of motion and balance, resulting in postural instability and exacerbating the overall discomfort experienced [1, 13].

Mixed sickness poses a major obstacle to adopting and developing VR platforms as a leisure activity in autonomous vehicles. Symptoms such as headaches, dizziness, and disorientation [4, 18, 22] can disrupt users' focus on VR content [1, 23] and negatively impact task performance [9, 17]. Addressing these challenges is imperative to fully realize VR platforms' potential as an integrated component of the autonomous vehicle experience.

To this end, researchers have investigated strategies to mitigate mixed sickness by enhancing the sense of motion during in-vehicle VR experience. One common approach employs visual feedback mechanisms, implemented as additional VR imagery integrated into VR content, to simulate movement and speed in alignment with the direction of travel. This is to align the user's vestibular sense of one's (vehicle) motion to the visual thereby reducing the sensory mismatch and ensuing sickness. Examples of such VR imagery include particle flow [3, 7, 9, 17], maps [9], and 3D grid patterns [18], all of which aim to strengthen users' sense of motion [1]. Another

approach is to provide haptic feedback, e.g. delivered through vehicle seats, such as vibrations and shaking synchronized with the vehicle's motion. Early systems like COMS-VR focused on front-back translational motion [12], while more advanced systems now incorporate tilting movements in all directions [1, 24]. Note that these visual and haptic feedback methods not only reduce sickness but also enhance user enjoyment [12], presence [4, 7], and comfort [1, 24] during in-vehicle VR experiences.

As outlined, as individual modality feedback has shown effectiveness on different causes of cyber or motion sickness and aspects of user experience, a natural extension would be to investigate the combined effect. One cannot simply assume that the combined approach would result in further extended sickness reduction and improved user comfort. For instance, it is not clear whether haptic feedback in the direction of the vehicle motion, but opposite to the body's inertia (in acceleration/deceleration) may interfere with the attempt to reduce sensory mismatch by the synchronized visual motion. While some research [4, 20] has investigated the multimodal feedback approach for sickness reduction, it has not been sufficiently explored.

In this paper, we present the development of an in-vehicle VR system that simultaneously delivers supplemental visual and haptic feedback to mitigate cyber and motion sickness during usage. The visual feedback features flying particles and a 3D rotating to reflect and visualize the vehicle's motion, synchronizing the visual and vestibular movement, while the haptic feedback employs a tiltable seat to alleviate the impact of acceleration stimuli, another cause for sickness. A pilot study was conducted to explore the following research questions (RQ), but not limited to.

- RQ1.** What is the most effective method for mitigating the mixed (cyber and motion) sickness during VR experiences in autonomous vehicles: haptic feedback vs. visual feedback vs. combined?
- RQ2.** How does the combination of haptic and visual feedback influence the in-vehicle VR experience?

This paper is organized as follows: Section 2 outlines the study design, followed by the experimental results in Section 3. Section 4 discusses the key findings, and Section 5 concludes the paper.

2 PILOT EXPERIMENT**2.1 Experimental Set-up and Materials**

This section describes the test on-vehicle VR system that provides visual and haptic feedback to mitigate mixed sickness.

2.1.1 Self-driving Car Trajectory and Control. For better experimental controllability, the validation experiment was conducted using a programmable autonomous vehicle (called XR-Cabin). In addition, the experiment was conducted in a designated self-driving car testing space (parking lot with no pedestrians nor signal lights). The space featured a total area of approximately 22 meters (width) × 122 meters (depth), with an average slope of 0.47 degrees. The area was divided into three distinct sections, each measuring approximately 22 meters in width and 26 meters in depth. These sections were utilized for vehicle movement, including trajectory navigation and turning. In the third section, the track had a slope of about 5 degrees, introducing slight curvature to the self-driving route.

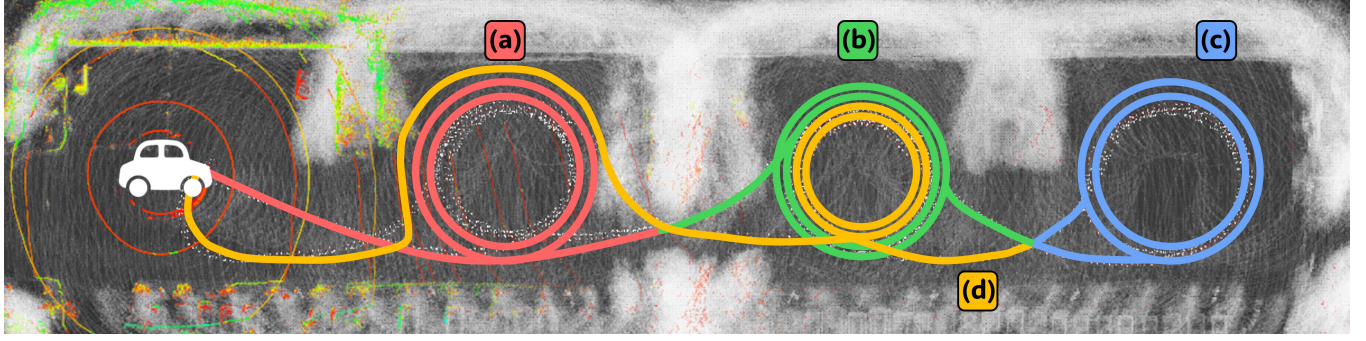


Figure 2: Driving trajectory for the autonomous vehicle in point cloud data. The total route length was approximately 616 meters, with each trial lasting about 4 minutes. The self-driving route was divided into four segments: (a) in the first segment, the vehicle made two consecutive left turns; (b) in the second, it made two right turns; (c) in the third, it repeated two left turns; and (d) in the fourth, it made two right turns followed by a large circular turn to return to the starting position.

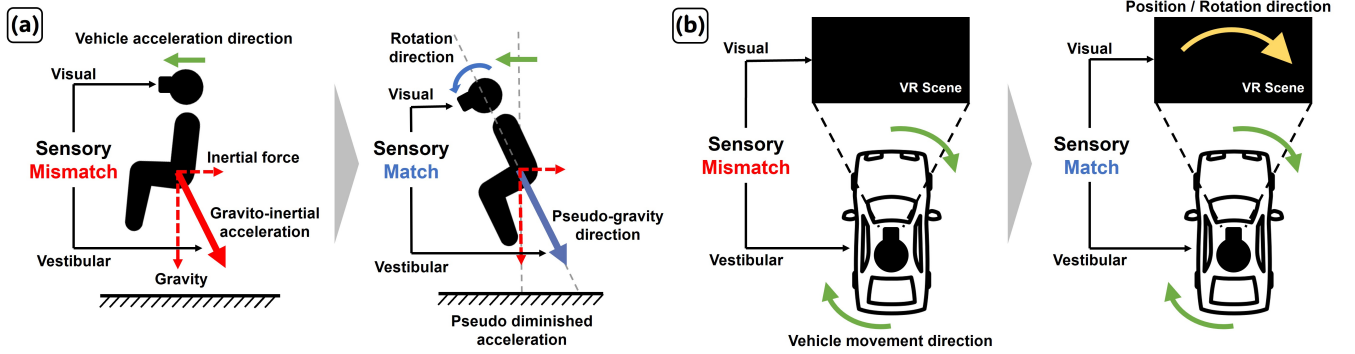


Figure 3: Illustration of the (a) haptic feedback [1] and (b) visual feedback [18] for alleviating the cyber and motion sickness.

The total length of the route, by which the autonomous car was programmed to run, was about 616 meters, and the car operated for about 4 minutes per trial. Within this duration, the vehicle performed acceleration, deceleration, and turning rotation. The vehicle's speed was set to specific values at each position along the route, ranging between 9–13 km/h. The self-driving route (see Figure 2) was divided into four segments: in the first segment, the vehicle made two consecutive left turns; in the second, it made two right turns; in the third, it repeated two left turns; and in the fourth, it made two right turns followed by a large circular turn to return to the starting position. This trajectory was designed to provide sufficient conditions for inducing motion sickness in participants.

2.1.2 Haptic System - Tiltable Seat. The haptic system reduces the acceleration stimuli from the vehicle, aligning the stationary visual information from the main VR content with the vestibular sensations. This alignment minimizes the sensory mismatch, thereby reducing the sickness and alleviating discomfort for passengers.

The reduction of acceleration stimuli is achieved through the tiltable seat [1]. As shown in Figure 3 (a), when inertial force is applied to the passenger from the vehicle, one perceives the resultant vector of this inertial force and gravity as the apparent acceleration stimulus, known as the Gravito-Inertial Acceleration (GIA) [2].

By tilting the seat such that the GIA aligns vertically downward relative to the passenger, the acceleration stimulus from the vehicle becomes imperceptible to them, leading to the illusion of being stationary. At this time, the seat's tilting speed is set to ensure the motion itself remains unnoticed. In addition, by wearing a VR headset, the tilting of the seat cannot be noticed even visually.

The vehicle's acceleration is either detected by the on-board sensor or through the pre-programmed motion profile of the autonomous test vehicle.

2.1.3 Visual System - Reflecting Linear and Circular Motion of Vehicle. The visual feedback [18] operates by sensing the linear and rotational vehicle motion (OBD II and IMU sensors). The main VR content would be surrounded by a large spherical grid (referred to as the Visual Grid), providing a consistent visual frame of reference for the user. The Visual Grid was designed to rotate in synchronization with the sensed real-world vehicle's rotation, including yaw, pitch, and roll motions. This rotational motion provided users with visual feedback corresponding to the vehicle's dynamics (see Figure 3 (b)). The grid had a diameter of approximately 30 meters, ensuring it encompassed the entire virtual environment and remained visible in the user's peripheral vision (about 50° in diameter).

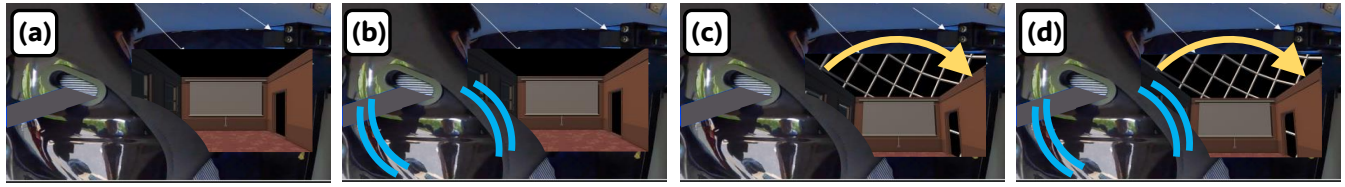


Figure 4: Compared test conditions by the type of feedback for alleviating the sickness and effects toward user experience: (a) No feedback (as baseline condition), (b) haptic feedback only, (c) visual feedback only, and (d) combined feedback.

The lateral motion of the vehicle (also sensed) was visualized by a particle system (flying cubes). These cube particles emerged in the user's forward direction, moving in synchronization with the vehicle's motion data and speed. The system was configured to spawn 55.58 particles per second to replace up to 1000 cubes. Particle parameters, such as size and lifetime, were set based on previous research [18].

Note that both the particles and the grid are seen as situated in the background and visual periphery, so as not to interfere with the main static content in the center. Nevertheless, based on the fact that reflected and synchronized motion is well detected in the visual periphery [17, 18].

2.1.4 Static Test VR Content. The VR test content was configured to be non-dynamic, with no user-initiated motion to minimize external bias. The experiment utilized Meta Quest 3 [16] to create a virtual office environment where the participant was seated. The virtual office environment featured windows on the sides, with an open ceiling allowing the participant to see a black background at the rear. Within this background, a visual grid and particles appeared, which the participant could perceive. In this environment, the participant performed the task of watching a music video (titled "Pretender" by the artist, "Official HIGE DANDISM" [19]), displayed on a virtual 2D screen positioned about 2.5 meters in front of the participant.

The dimensions of the virtual space were set to about 3m (width) $\times$ 4m (length) $\times$ 3m (height) to provide a realistic yet controlled setting. A virtual screen was positioned about 2.5m in front of the participant's seated position, where the music video was displayed. The music video had a runtime of approximately 5 minutes, slightly exceeding the actual vehicle travel time of 4 minutes. To ensure the experiment aligned with the travel duration, the system was configured to automatically terminate if the experiment was interrupted during the route. This 4-minute duration was deemed sufficient to elicit a reasonable level of cybersickness in participants [10].

2.2 Experiment Design and Hypotheses

The experiment followed a 1×4 within-subject design to compare the levels of sickness reduction by each feedback and their combinations (see Figure 4). Participants experienced the following treatments in a counter-balanced order.

- **Baseline (No feedback):** Participants rode in an autonomous vehicle and viewed the main static VR content without any haptic feedback or vehicle motion reflecting visual feedback (particles/grid).

- **Haptic feedback only:** Participants rode in an autonomous vehicle and viewed the main static VR content with only the haptic (seat tilt by acceleration) feedback.
- **Visual feedback only:** Participants rode in an autonomous vehicle and viewed the main static VR content with only the supplemental visual feedback, namely the particles and rotating grid reflecting the vehicle motion.
- **Combined (Haptic + Visual) feedback:** Participants rode in an autonomous vehicle and viewed the main static VR content with both visual and haptic feedback.

Based on the literature review, three hypotheses were proposed.

- **H1:** The tiltable seat in the haptic feedback system is expected to reduce pressure caused by vehicle movement, improve postural stability, and alleviate sensory mismatch, leading to improved comfort.
- **H2:** The linear and circular visualization of the user/vehicle motion will enhance the user's sense of movement, thereby reducing sensory mismatch during the VR experience.
- **H3:** The haptic and visual feedback systems will contribute to a relatively larger reduction in the sickness compared to providing individual modality feedback by reduced sensory mismatch and improved comfort.

2.3 Dependent Variables

Subjective variables were collected to examine the effects of each feedback modality and their combination on mitigating mixed sickness and enhancing overall user experience. After completing each treatment, participants responded to the following questionnaires on a laptop. All questionnaires can be found in the Appendix.

- **Level of Sickness (SSQ):** Kennedy's simulator sickness questionnaire (SSQ) [11] was used to measure the level of mixed sickness. SSQ consisted of reporting 16 certain symptoms related to sickness (e.g., Headache, Dizziness, and Fatigue) on a 4-point Likert scale (0: None, 3: Severe). Following the scoring mechanism, we could derive *Total* scores for the level of cybersickness and as related to *Nausea*, *Oculomotor*, and *Disorientation* scores [11].
- **Sense of Movement (SM):** This refers to the user's subjective perception of motion while using VR content in the autonomous vehicle. A total of 9 questions were prepared to measure key aspects of *Perceived Movement* (4 questions), including the perception of speed, acceleration, direction, and vibration; the *Consistency* between visual and physical senses of movement (4 questions); and lastly, the *Influence* of movement sensation on mixed sickness, as subjectively

judged by participants (1 question); These questions were designed based on prior research on motion perception in dynamic environments [27, 29]. Participants rated their responses on a 7-point Likert scale (1: Not at all, 7: Very Much), and averages were calculated for three main factors.

- **Level of Comfort:** Each device used in the experiment was considered to have an effect on alleviating discomfort caused by visual strain and postural instability. To evaluate this, *Visual Comfort*, *Postural Comfort*, and *Overall Comfort* were each assessed using a single item on a 7-point Likert scale (1: Not at all, 7: Very Much).
- **Presence:** This refers to the user's sense of "being there" in a virtual environment, known to be influenced by cybersickness in VR [10]. To measure this, three specific items (1, 3, and 6) from the Slater-Usuh-Steed questionnaire [30] were selected to reduce participant fatigue. Each item was rated on a 7-point Likert scale (1: Not at all, 7: Very Much), and the average score of these items was calculated.

2.4 Participants

Initially, 12 participants were recruited from the university; however, due to technical issues during the experiment, data from only 8 participants (3 females and 5 males) were used for analysis. Their ages ranged from 22 to 28 ($M = 24.88$; $SD = 1.96$). All participants had normal hearing and normal/corrected vision, with 2 participants wearing glasses. Their VR familiarity was checked using a 7-point Likert scale (1: not at all, 7: very much), resulting in an average of 3.33 ($SD = 1.51$). Participants also reported their susceptibility to motion sickness using the MSSQ [6], with the results showing an average of 7.77 ($SD = 6.64$). This study was approved by the Ethics Committee of the Nara Institute of Science and Technology, the ethical no. 2021-I-12-1, and was conducted according to institutional ethical provisions and the Declaration of Helsinki.

2.5 Experimental Procedure

When participants arrived at the experimental site for self-driving, the experimenter briefly explained the experimental procedure to the participants and received a consent form. Afterward, a pre-experiment survey was given to record and collect the level of familiarity with VR, susceptibility to motion sickness (MSSQ) [6], and demographic information.

With the experimenter's assistance, participants were seated in the vehicle seat and fastened their seatbelts. Then, they wore the VR headset (Meta Quest 3), which was calibrated to their position. Once the experiment began, the autonomous vehicle followed a predetermined driving route while participants viewed the assigned VR content. The driving route took about 4 minutes to complete, meaning participants experienced each condition for 4 minutes.

At the end of each condition, participants removed the VR headset and exited the vehicle to complete the post-condition survey on a laptop. They were then provided with a break of about 10 minutes (or more, if needed). Meanwhile, at the end of each treatment, the experimenter reset the seat's tilting position to its initial state.

The experimental order was counterbalanced using a Balanced Latin Square design to mitigate potential order effects. Upon completion of all treatments, participants were asked to complete a

post-experiment survey followed by a brief interview. The entire experiment had a duration of approximately one hour.

3 RESULT AND DISCUSSION

This experiment utilized a within-subject design with four treatments, requiring a one-way repeated measures analysis for statistical evaluation. Due to the small number of participants, the Friedman test, a non-parametric method for repeated measures, was used. Post-hoc analysis was conducted using the Conover test with Bonferroni correction. The significance level for all statistical analyses was set at 5%. The statistical results are summarized in Figure 5 and Table 1.

3.1 Level of Comfort

The comfort questionnaire was designed to assess the sources of comfort experienced during the VR experience by categorizing responses into visual, haptic, and overall comfort levels. Among these, only the comfort associated with haptic feedback showed a statistically significant difference (see Figure 5 (c)). Specifically, haptic feedback resulted in improved comfort compared to the baseline condition (Baseline < Haptic; $p = .016^*$), and an even higher level of comfort was observed when haptic and visual feedback were combined (Baseline < Combined; $p < .001^{***}$). This outcome suggests that the tiltable seat, which was synchronized with the vehicle's movements but operated in the opposite of inertial force, may have diminished the pressure exerted by the vehicle's motion, allowing participants to perceive reduced pressure. As a result, comfort levels were relatively higher compared to other conditions. This reduction in perceived pressure may have contributed to improved postural stability, reduced the frequency and magnitude of sensory mismatches, and ultimately alleviated mixed sickness. These findings support the first hypothesis (H1).

3.2 Sense of Movement

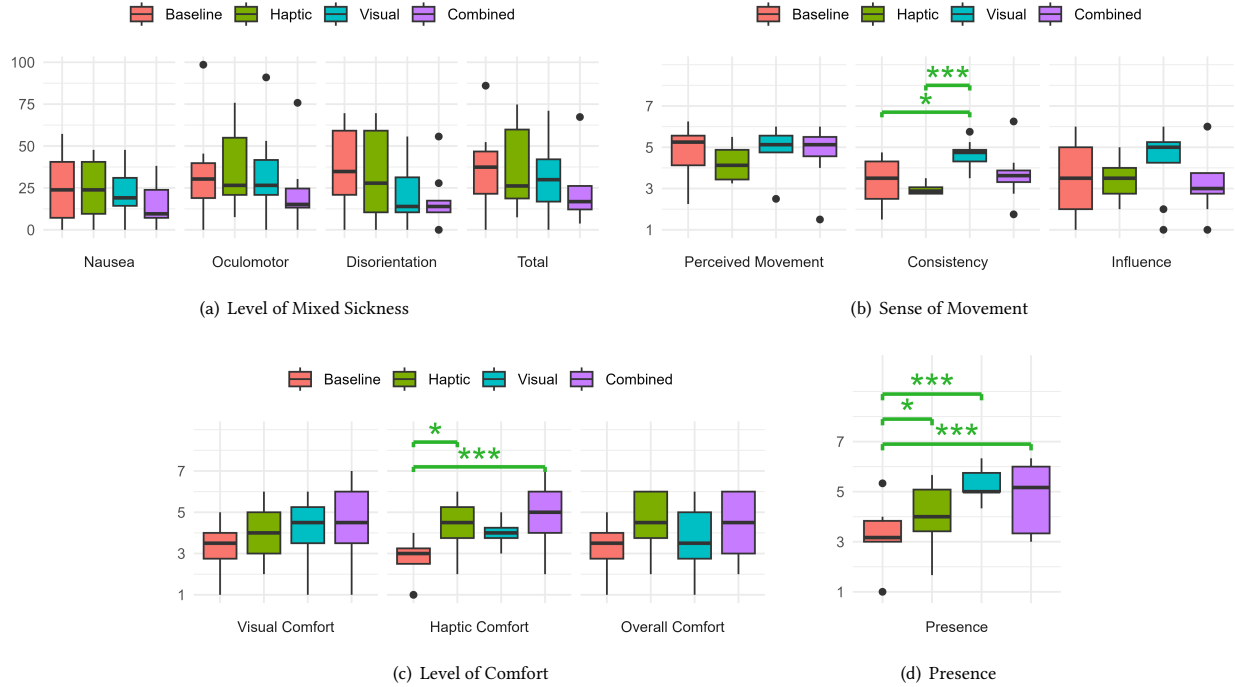
The sense of movement questionnaire was analyzed based on three factors: perceived movement, consistency, and influence. Among these, a statistically significant difference was found only in the consistency factor (see Figure 5 (b)). Specifically, visual feedback resulted in higher consistency scores compared to the baseline condition ($p = .0106^*$) and was notably higher than haptic feedback ($p < .001^{***}$). This suggests that the visual grid pattern and particle effects, which were synchronized with the vehicle's movements, enhanced participants' sense of movement and helped them better perceive their direction of travel. As a result, the visual feedback likely contributed to reducing sensory mismatch and alleviating mixed sickness. These findings support the second hypothesis (H2).

3.3 Level of Mixed Sickness

The assessment of mixed sickness levels was conducted using the SSQ, with analyses performed based on scores for three key symptoms—nausea, oculomotor discomfort, and disorientation—along with the total score. However, no statistically significant differences were observed across conditions, and thus, the third hypothesis (H3) was not supported. Nevertheless, a decreasing trend in mixed sickness levels was observed across all measures when feedback was applied compared to the baseline condition, with even lower

Table 1: The results of statistical analysis (* $p < .05$; ** $p < .01$; and * $p < .001$). Due to the limited number of participants, the Friedman test was conducted, followed by the Conover test for pairwise comparisons (post-hoc) with Bonferroni correction.**

		Mean (SD)				Friedman test			post-hoc
		Baseline	Haptic	Visual	Combined	χ^2	ES	p	
Mixed Sickness	Nausea	25.0 (21.6)	25.0 (18.3)	21.5 (16.7)	15.5 (15.2)	3.75	0.156	.290	
	Oculomotor	34.1 (30.0)	36.0 (24.2)	34.1 (27.8)	22.7 (23.3)	3.73	0.155	.292	
	Disorientation	36.5 (27.8)	33.1 (29.7)	20.9 (19.7)	17.4 (17.8)	5.33	0.222	.149	
	Total	36.5 (27.0)	36.5 (24.7)	30.9 (22.4)	22.0 (20.2)	4.01	0.167	.260	
Sense of Movement	Perceived Movement	4.72 (1.40)	4.22 (0.88)	4.94 (1.08)	4.69 (1.42)	3.04	0.127	.386	
	Consistency	3.38 (1.16)	2.97 (0.28)	4.62 (0.73)	3.69 (1.29)	10.6	0.443	.013*	Baseline < Visual*; Haptic < Visual***
	Influence	3.50 (1.77)	3.38 (1.06)	4.38 (1.85)	3.38 (1.77)	2.10	0.087	.551	
Level of Comfort	Visual	3.25 (1.28)	4.00 (1.31)	4.12 (1.81)	4.38 (2.07)	3.09	0.129	.378	
	Haptic	2.75 (1.16)	4.38 (1.41)	4.00 (0.75)	4.88 (1.55)	10.3	0.430	.016*	Baseline < Combined***; Baseline < Haptic*
	Overall	3.25 (1.28)	4.50 (1.51)	3.62 (1.69)	4.38 (1.60)	4.23	0.176	.238	
Presence		3.28 (1.42)	4.00 (1.46)	5.28 (0.74)	4.78 (1.54)	12.2	0.675	.006**	Baseline < Combined***; Baseline < Visual***; Baseline < Haptic*

**Figure 5: Boxplots indicating statistical results: (a) level of mixed sickness, (b) sense of movement, (c) level of comfort, and (d) presence (statistical significant levels: * $p < .05$, ** $p < .01$, and *** $p < .001$).**

levels recorded when both feedback modalities were combined (see Figure 5 (a)). The lack of significant differences in sickness is largely due to the stationary and passive nature of the VR content, which involved watching a music video, while the limited number of participants may have also influenced the results. Therefore, future studies will explore the effects of feedback in more dynamic and active VR environments with a larger sample size to better understand the differences across treatments.

As a partial answer to RQ1, the findings suggest that haptic feedback may contribute to reducing mixed sickness by improving comfort, while visual feedback may help alleviate mixed sickness by enhancing the sense of movement. When combined, these feedback modalities could potentially lead to a greater reduction in mixed sickness. However, this pilot study did not determine the superiority of one feedback type over the other but only identified the possibility that their combination may result in a greater reduction.

3.4 Level of Presence

To examine the impact of each feedback modality on users' in-vehicle VR experience, the presence questionnaire [30] was utilized. The analysis results (see Figure 5 (d)) indicated statistically significant improvements across all feedback conditions compared to the baseline: Haptic ($p = .018^*$), Visual ($p < .001^{***}$), and Combined ($p < .001^{***}$). These improvements are attributed to the enhancements in comfort and reductions in sensory mismatch provided by the feedback modalities, which likely mitigated the effects of sickness and contributed to an increased sense of presence.

In response to RQ2, the findings suggest that each feedback modality can enhance presence, a key factor in VR user perception, and their combination also contributes to an improved presence.

4 CONCLUSION

In this paper, different approaches to mitigating cyber and car motion sickness experienced while using VR content in the autonomous vehicle were evaluated and compared. The results showed that visual feedback significantly enhanced the sense of movement, while haptic feedback significantly improved user comfort during the VR experience. These enhancements may have contributed to a reduction in sickness levels, with a trend indicating lower sickness levels when both feedback modalities were combined. However, the differences in sickness levels across treatments were not statistically significant, possibly due to the limited number of participants or the low sickness levels induced by the stationary and passive nature of the VR content used (watching a music video). Nevertheless, the observed trend of decreasing sickness with the addition of feedback suggests the need for further investigation with a larger participant pool and more dynamic VR content in future research.

Acknowledgments

This research was supported by the MSIT, Korea, under the Global Research Support Program in the Digital Field program (RS-2024-00431357) supervised by the IITP.

References

- [1] Takuto Akiyoshi, Yuki Shimizu, Yusaku Takahama, Koki Nagata, and Taishi Sawabe. 2023. Hype D-Live: XR Live Music System to Entertain Passengers for Anxiety Reduction in Autonomous Vehicles. In *2023 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. IEEE, 148–156.
- [2] Houshyar Asadi, Shady Mohamed, Chee Peng Lim, and Saeid Nahavandi. 2016. A review on otolith models in human perception. *Behavioural brain research* 309 (2016), 67–76.
- [3] Hyung-Jun Cho and Gerard J Kim. 2022. RideVR: reducing sickness for in-car virtual reality by mixed-in presentation of motion flow information. *IEEE Access* 10 (2022), 34003–34011.
- [4] Ahmed Ibrahim Ahmed Mohamed Elsharkawy, Aya Abdunnasser Saed Ataya, Dohyeon Yeo, Eunsol An, Seokhyun Hwang, and SeungJun Kim. 2024. SYNC-VR: Synchronizing Your Senses to Conquer Motion Sickness for Enriching In-Vehicle Virtual Reality. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–17.
- [5] Bocheon Gim, Seongjun Kang, Gwangbin Kim, Dohyeon Yeo, Seokhyun Hwang, and Seungjun Kim. 2024. Curving the Virtual Route: Applying Redirected Steering Gains for Active Locomotion in In-Car VR. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–7.
- [6] John F Golding. 1998. Motion sickness susceptibility questionnaire revised and its relationship to other forms of sickness. *Brain research bulletin* 47, 5 (1998).
- [7] Philipp Hock, Sebastian Benedikter, Jan Gugenheimer, and Enrico Rukzio. 2017. Carvr: Enabling in-car virtual reality entertainment. In *Proceedings of the 2017 CHI conference on human factors in computing systems*. 4034–4044.
- [8] Hololide. 2021. Adding thrill to every ride — conceptual litepaper v.2.1. In *hololide* (2021), 1–26.
- [9] Hwaseung Jeon, Taewoo Jo, Dohyeon Yeo, Eunsol An, Yumin Kang, and SeungJun Kim. 2024. The way of water: exploring the role of interaction elements in usability challenges with in-car VR experience. *Virtual Reality* 28, 3 (2024), 121.
- [10] Dongyun Joo, Hanseob Kim, and Gerard Jounghyun Kim. 2024. The Effects of False but Stable Heart Rate Feedback on Cybersickness and User Experience in Virtual Reality. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–15.
- [11] Robert S Kennedy, Norman E Lane, Kevin S Berbaum, and Michael G Lillenthal. 1993. Simulator sickness questionnaire: An enhanced method for quantifying simulator sickness. *The international journal of aviation psychology* 3, 3 (1993).
- [12] Ryo Kodama, Masahiro Koge, Shun Taguchi, and Hiroyuki Kajimoto. 2017. COMS-VR: Mobile virtual reality entertainment system using electric car and head-mounted display. In *2017 IEEE symposium on 3D user interfaces*. IEEE.
- [13] Joseph J LaViola Jr. 2000. A discussion of cybersickness in virtual environments. *ACM Sigchi Bulletin* 32, 1 (2000), 47–56.
- [14] Mark McGill, Alexander Ng, and Stephen Brewster. 2017. I Am The Passenger: How Visual Motion Cues Can Influence Sickness For In-Car VR. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems* (Denver, Colorado, USA) (CHI '17). Association for Computing Machinery, New York, NY, USA, 5655–5668. <https://doi.org/10.1145/3025453.3026046>
- [15] Mark McGill, Graham Wilson, Daniel Medeiros, and Stephen Anthony Brewster. 2022. Passengxr: A low cost platform for any-car, multi-user, motion-based passenger xr experiences. In *Proceedings of the 35th annual ACM symposium on user interface software and technology*. 1–15.
- [16] Meta. 2025. Meta Quest 3: Advanced All-In-One VR Headset. <https://www.meta.com/quest-3/>
- [17] Sewon Noh, Junseo Park, and Gerard J Kim. 2024. Effect of Synchronized Vehicle Motion to Mitigating Sickness for On-Vehicle VR during Active Task Performance. *IEEE Access* (2024).
- [18] Sewon Noh, Seunghoon Park, and Gerard J Kim. 2024. OnVehicleVR: Mitigating Sickness in On-Vehicle Virtual Reality by Mixing in Synchronized Vehicle Motion Information. *IEEE Access* 12 (2024), 148374–148386.
- [19] Official HIGE DANDISM. 2019. Pretender (Official Video). <https://youtu.be/TQ8WIA2GXbk>
- [20] Katharina Margareta Theresa Pöhlmann, Gang Li, Mark McGill, Reuben Markoff, and Stephen Anthony Brewster. 2023. You spin me right round, baby, right round: Examining the impact of multi-sensory self-motion cues on motion sickness during a VR reading task. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–16.
- [21] Zhanyan Qiu, Mark McGill, Katharina Margareta Theresa Pöhlmann, and Stephen Anthony Brewster. 2024. Acceleration instead of Speed: Acceleration Visual Cues in VR for Reduced Motion Sickness in Linear Motion. In *Adjunct Proceedings of the 16th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*. 56–61.
- [22] Yusuke Sakai, Toshimitsu Watanabe, Yoshio Ishiguro, Takanori Nishino, and Kazuya Takeda. 2019. Effects on user perception of a 'modified' speed experience through in-vehicle virtual reality. In *Proceedings of the 11th International Conference on Automotive User Interfaces and Interactive Vehicular Applications: Adjunct Proceedings*. 166–170.
- [23] Taishi Sawabe, Masayuki Kanbara, Norimichi Ukita, Tetsushi Ikeda, Luis Yoichi Morales Saiki, Atsushi Watanabe, and Norihiro Hagita. 2015. Comfortable autonomous navigation based on collision prediction in blind occluded regions. In *IEEE International Conference on Vehicular Electronics and Safety*. IEEE, 75–80.
- [24] Taishi Sawabe, Yosuke Okami, Masayuki Kanbara, Yuichiro Fujimoto, and Hirokazu Kato. 2022. Diminished Reality for Sense of Movement with XR Mobility Platform. In *2022 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*. IEEE, 348–351.
- [25] Alexander Schäfer, Gerd Reis, and Didier Stricker. 2022. A survey on synchronous augmented, virtual, and mixed reality remote collaboration systems. *Comput. Surveys* 55, 6 (2022), 1–27.
- [26] Tianyu Shen, Shi-Sheng Huang, Deqi Li, Zhiyuan Lu, Fei-Yue Wang, and Hua Huang. 2022. VirtualClassroom: A lecturer-centered consumer-grade immersive teaching system in cyber-physical-social space. *IEEE Transactions on Systems, Man, and Cybernetics: Systems* 53, 6 (2022), 3501–3513.
- [27] Yuki Shimizu, Taishi Sawabe, Masayuki Kanbara, Yuichiro Fujimoto, and Hirokazu Kato. 2024. Development of Comfort Environment in Autonomous Driving by Reducing the Passenger's Sense of Movement. *IEICE Technical Report; IEICE Tech. Rep.* 123, 380 (2024), 72–77.
- [28] Gunnar Stevens, Paul Bossauer, Stephanie Vonholdt, and Christina Pakusch. 2019. Using time and space efficiently in driverless cars: Findings of a co-design study. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*.
- [29] Thomas A Stoffregen and L James Smart Jr. 1998. Postural instability precedes motion sickness. *Brain research bulletin* 47, 5 (1998), 437–448.
- [30] Martin Usoh, Ernest Catena, Sima Arman, and Mel Slater. 2000. Using presence questionnaires in reality. *Presence* 9, 5 (2000), 497–503.
- [31] Ekim Yurtsever, Jacob Lambert, Alexander Carballo, and Kazuya Takeda. 2020. A survey of autonomous driving: Common practices and emerging technologies. *IEEE access* 8 (2020), 58443–58469.

A Questionnaire

No.	English	Korean (한국어)	Japanese (日本語)
1	Speed perception: How well did you perceive the speed of the vehicle?	속도 감각: 차량이 움직이는 속도를 얼마나 느꼈습니까?	速度感覚: 車両の速度をどの程度感じましたか?
2	Acceleration/Deceleration perception: How well did you recognize the acceleration or deceleration of the vehicle?	가속/감속 감각: 차량이 가속하거나 감속할 때 얼마나 인지했습니까?	加速/減速感覚: 車両が加速または減速する際、どの程度認識しましたか?
3	Direction perception: How well did you perceive the direction of the vehicle (e.g., left or right turns)?	방향 감각: 차량의 진행 방향(좌우회전 등)을 얼마나 느꼈습니까?	方向感覚: 車両の進行方向 (左右の回転など) をどの程度感じましたか?
4	Vibration perception: How strongly did you feel the vibrations or shaking of the vehicle?	진동 감각: 차량의 흔들림이나 진동을 얼마나 느꼈습니까?	振動感覚: 車両の振動や揺れをどの程度強く感じましたか?
5	Did the visual and physical sense of movement align?	시각과 체감의 이동이 일치했습니까?	視覚と体感の移動が一致しましたか?
6	Did you feel the movement of the vehicle excessively?	차량의 움직임을 과하게 느꼈습니까?	車両の動きを過剰に感じましたか?
7	How well could you predict how far the vehicle had actually moved?	실제로 얼마나 이동했는지 잘 예측할 수 있었습니까?	実際にどれだけ移動したのか、うまく予測できましたか?
8	Did you feel that the changes in the vehicle's movement were predictable?	차량의 이동 변화가 예상 가능하다고 느꼈습니까?	車両の移動変化が予測可能だと感じましたか?
9	To what extent did the intensity of the sense of movement affect your mixed sickness symptoms?	이동 감각의 강도가 멀미 증상에 얼마나 영향을 미쳤습니까?	移動感覚の強度が車酔い症状にどの程度影響しましたか?

Table 2: Questionnaire for Sense of Movement – The questionnaire consists of 9 items, where items 1 to 4 assess the perceived movement, items 5 to 8 evaluate the consistency between visual and physical senses of movement, and items 9 examine the influence on mixed sickness. All questions are rated on a 7-point Likert scale (1: Not at all; 7: Very much).

No.	English	Korean (한국어)	Japanese (日本語)
1	I had a sense of “being there” in the virtual environment.	가상환경에서 "거기에 있다"는 느낌이 들었습니다.	私はバーチャル環境で「そこにいる」という感覚を持ちましたか?
2	When you think back about your experience, do you think of the virtual environment more as images that you saw, or more as somewhere that you visited?	자신의 경험을 돌아볼 때 가상환경을 본 이미지로 생각하십니까, 아니면 방문한 곳(가상환경)으로 더 많이 생각하십니까?	自分の体験を振り返ったとき、そのバーチャル環境を「見た映像」としてより捉えましたか、それとも「実際に居る場所（仮想環境）」としてより捉えましたか?
3	During the time of the experience, did you often think to yourself that you were actually in the virtual environment?	체험기간 동안, 실제로 가상환경에 있다고 자주 생각했나요?	体験中に、自分が実際にバーチャル環境にいると思いましたか?

Table 3: Questionnaire for Presence. The original questionnaire [30] consisted of six items; however, to reduce participant fatigue, only three items were used. All items were rated on a 7-point Likert scale (1: Not at all, 7: Very much), and the final score was calculated by averaging the responses across the three items.

No.	English	Korean (한국어)	Japanese (日本語)
1	Did the visual information make the experience comfortable?	시각적인 정보 제공으로 인해 체험이 편했습니까?	視覚的な情報提供によって体験は快適になりましたか?
2	Did the movements of the vehicle or seat make the experience comfortable?	차량 및 의자의 움직임으로 인해 체험이 편했습니까?	車両や椅子の動きによって体験は快適になりましたか?
3	Overall, how comfortable was the experience and ride in terms?	전반적인 체험과 승차감이 편했습니까?	全体的な経験と乗り心地は快適でしたか?

Table 4: Questionnaire for the Level of Comfort. This questionnaire evaluates the comfort levels associated with different types of feedback through three items: comfort with visual feedback, comfort with haptic feedback, and overall comfort. Responses were collected using a 7-point Likert scale (1: Not at all; 7: Very Much). As the experiment included conditions where participants experienced either visual or haptic feedback separately, each item was analyzed independently.